

Two-term expansion for a polynomial perturbation: how the monomials of ξ populate $\Lambda(h, k)$

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2026-09-06

Abstract

The first step beyond a single monomial: for $\xi(u) = \xi_0 + \sum_{j < D} c_j u^{j+1}$ (any ξ analytic at 0, truncated) the first-order term of the one-dimensional Taylor tree is a *sum over the monomials of ξ* , each contributing its own exponent $(h+j+2)/(2k)$ with coefficient $\beta c_j \frac{1}{2k} S_{(h+j+2)/(2k)+1/2}(\xi_0)$, and the remainder is bounded by an explicit multiple of $n^{-(h+3)/(2k)}$ plus boundary tails. This exhibits the union structure of the candidate exponent set: distinct monomials of the perturbation land on distinct points of $\Lambda(h, k)$. Zero `sorry/axiom`.

1 Setting

Units 28–31 treat $\xi - \xi_0 = cu^m$. The paper's ξ is analytic with $\xi(0) = \xi_0$, so $J(u) = \xi(u) - \xi_0$ is a power series with no constant term; its truncation is a polynomial $\sum_{j < D} c_j u^{j+1}$. The second-order machinery of unit 28 goes through verbatim with $x = \beta\sqrt{n}u^k J(u)$ once one bound is available: on the chart $|J(u)| \leq Cu$ with $C = \sum_{j < D} |c_j| b^j$, which controls both x^2 (giving u^2 , hence the exponent shift $2/(2k)$) and $e^{|x|}$ (a population kernel at $\xi_0 + Cb$).

2 The thing formalised

```
theorem standardIntegral1D_polynomial_second_order ( n b : ) ( c : → )
  ( h k D : ) ( h : 0 < ) ( hn : 1 ≤ n ) ( hb : 0 < b ) ( hk : 0 < k ) :
  | Z_chart( + {j < D} c j u^(j+1)) - (1/(2k)) n^(-(h+1)/(2k)) S_{(h+1)/(2k)}( )
    - ∑_{j < D} c_j ((1/(2k)) n^(-(h+j+2)/(2k)) S_{(h+j+2)/(2k)+1/2}( )) |
  ( C )^2/2 ((1/(2k)) n^(-(h+3)/(2k)) S_{(h+3)/(2k)+1/2}( + C b))
  + tail + ∑_{j < D} | c_j | tail_{1,j}
```

3 Proof strategy

The pointwise decomposition $e^x = 1 + x + R(x)$ splits the integrand into $g_0 + g_1 + g_2$ exactly as in unit 28; the new element is $g_1 = u^h e^{\text{pop}} \cdot \beta\sqrt{n}u^k \sum_j c_j u^{j+1}$, which `Finset.mul_sum` turns into a finite sum of first-order insertion integrands, integrated by `integral_finsetSum` and evaluated termwise by the $p = 1$ insertion identity (half-line minus tail). For the remainder, $|J(u)| \leq Cu$ follows from `Finset.abs_sum_le_sum_abs` and $|c_j| u^{j+1} = |c_j| u^j \cdot u \leq |c_j| b^j \cdot u$; then $x^2 \leq (\beta\sqrt{n}u^k Cu)^2$ and $e^{|x|} \leq e^{\beta\sqrt{n}u^k Cb}$, the latter merging with the population kernel into the kernel at $\xi_0 + Cb$. The remainder integral is the $p = 2$ insertion at $h + 2$, so its n -power is $n^{-(h+3)/(2k)}$ with no manual `rpow` arithmetic. The final rearrangement is again a universally quantified `key` identity with five real slots.

4 Roadblocks and resolutions

One: the remainder bound `hI` was stated with the opaque constants a' and C while the goal carries them unfolded; `rw [ha', hC]` at `hI` aligns them. The insertion identity at $p = 2$ returns the order shift as $\uparrow 2/2$, normalised by `show ((2:):)/2 = 1` by `norm_num` inside the value computation rather than by a global `norm_num` (which would also rewrite the natural-number casts in the statement).

5 What was Mathlib, what was new

Mathlib: `Finset.mul_sum`, `integral_finsetSum`, `Finset.abs_sum_le_sum_abs`, `Finset.sum_le_sum`, `continuous_finsetSum`. New: the polynomial-perturbation two-term expansion and the chart bound $|J(u)| \leq Cu$ for a polynomial without constant term.

6 Lessons

The step from one monomial to a polynomial costs one inequality ($|J(u)| \leq Cu$) and one `Finset.mul_sum`; everything else is reuse. The same pattern should carry the all-orders theorem to polynomial ξ once the tree coefficients $\xi_{\mathbf{n},p}$ (multinomial expansion of J^p) are named.

7 Follow-ups

Higher orders for polynomial ξ (the one-dimensional `eq:flucttreeterms`), the O -packaging of this unit, and the multivariate tree.